

ADVANCED MACHINE LEARNING · WORKSHEET 13

Time Series Analysis

Stationarity · Differencing · AR, ACF/PACF and MA

TEACHER COPY — Lecture 13

PART I: WHY TIME MATTERS

In every model we have built so far — from Linear Regression to Regularization — we assumed that the order of rows in our dataset does not matter. Time Series Analysis begins where that assumption breaks down.

When the order of data matters

▷ HOOK

Consider two datasets. The first contains 500 house listings with features like area, bedrooms, and location. You randomly shuffle the rows before training a Linear Regression model. The predictions remain identical. Now consider a second dataset: daily server traffic on your university network recorded every hour for a year. You shuffle the rows. What happens to the data?

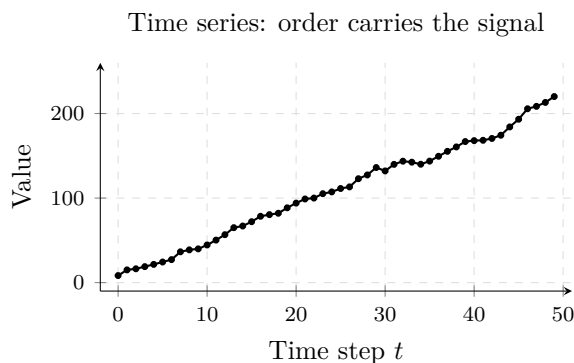
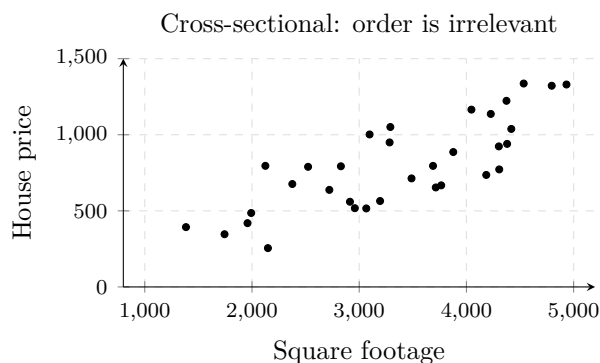
Independent observations vs. temporal sequences

In classical Machine Learning, we assume observations are **Independent and Identically Distributed (I.I.D.)**. The order of rows does not matter. Row 1 (a house in Delhi) is completely independent of Row 2 (a house in Mumbai). Each data point is an isolated, standalone event.

Classroom question: If you take a dataset of daily stock prices and completely randomize the order of the rows, what happens to the data?

Expected answer: It becomes meaningless noise. The trend is destroyed. The temporal dependencies between consecutive days are broken, and the data loses its forecasting value. Stock prices on consecutive days are not independent — today's price is heavily influenced by yesterday's.

For temporal data, the order of the rows is not an arbitrary index — it is a vital piece of information. The fact that row 5 happened immediately after row 4 is the most important feature of the dataset.



Shuffling the rows on the left changes nothing. Shuffling the rows on the right destroys the information the model needs.

★ KEY INSIGHT

Time Series Analysis is a branch of Machine Learning where time is the primary dimension and the past directly influences the future. Shuffling the data destroys its meaning — unlike cross-sectional data where row order is irrelevant.

What is a time series?

A **Time Series** is a sequence of data points recorded at successive, **equally spaced** intervals in time. Key structural differences from cross-sectional data:

Aspect	Cross-sectional data	Time Series data
Row index	Arbitrary ID	Timestamp
Row independence	Independent (I.I.D.)	Dependent (current row depends on previous rows)
Standard K-Fold CV	Valid	Invalid (causes data leakage)

Why can we not use standard K-Fold cross-validation for time series?

If you randomly split a time series into training and testing sets, you might end up training on data from 2026 to predict data from 2024. This violates the **arrow of time** and creates **data leakage**.

Expected answer: Expected Reasoning: Students should recall that K-Fold randomly assigns observations to folds, which would mix future data into the training set and past data into the test set.

Expected answer: Common Misconception: Students may think K-Fold is always valid. Emphasize that temporal ordering must be preserved — training data must come before test data in time.

Expected answer: Marking Hint: Accept any answer that mentions the violation of temporal ordering and the risk of using future data to predict the past.

◆ TAKEAWAY

Time series data replaces the concept of an arbitrary row ID with a specific timestamp. Because the current row depends on previous rows, standard cross-validation techniques cannot be applied directly. Shuffling temporal data destroys its structure.

Components of a time series

▷ HOOK

Imagine plotting the electricity consumption of your university campus over the past five years. You would likely see usage drop on weekends, drop during summer holidays, peak during exam weeks, and yet show an overall upward trend as more computers and servers are added each year. The observed line on the graph is not one chaotic pattern — it is actually several distinct patterns stacked on top of each other.

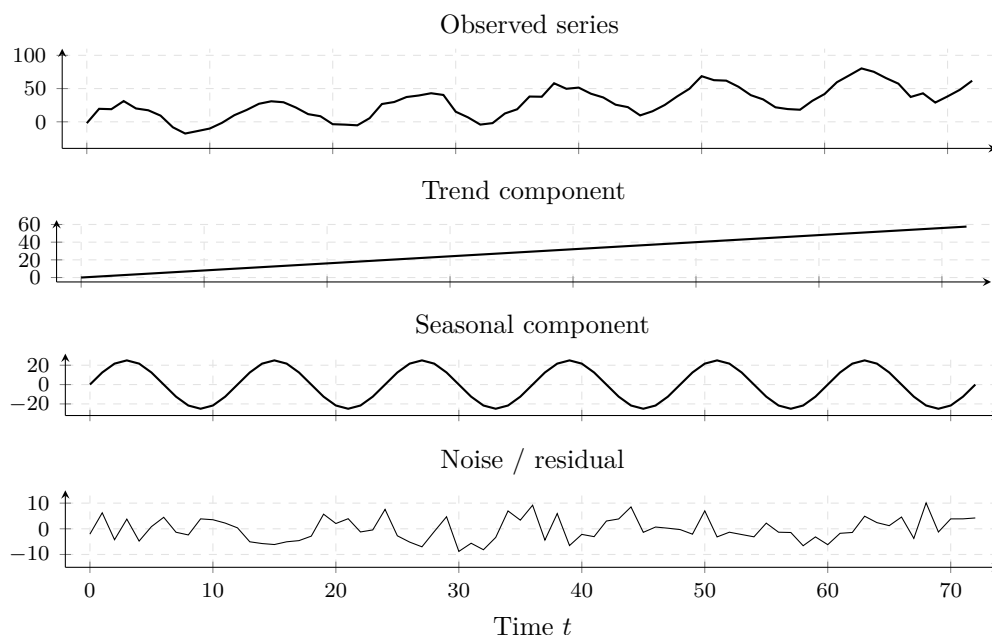
Decomposing a time series

An observed time series can be mathematically decomposed into three main components:

1. **Trend:** The overall, long-term direction of the data. Ignoring day-to-day bumps, is the general trajectory moving **upward** or **downward**?

2. Seasonality: Short-term, repeating cycles tied to specific timeframes. These repeat at known, **fixed** frequencies.

3. Noise (Residual / Random Shocks): The **random**, unpredictable variations that cannot be explained by trend or seasonality.



Classroom question: If a model's prediction errors (residuals) look exactly like pure random noise — no trend, no repeating pattern — what does that tell us about the model?

Expected answer: The model has perfectly captured everything that can be learned from the data. It has extracted all the trend and all the seasonality. The only thing remaining is pure, unpredictable noise — which no model can learn. This is the best possible outcome.

★ KEY INSIGHT

When we build forecasting models, the algorithm tries to extract the rules of the trend and the cycles of the seasonality. If the model's residuals are pure random noise, the model has captured all systematic information in the data.

◆ TAKEAWAY

Every time series can be decomposed into Trend (long-term direction), Seasonality (repeating cycles), and Noise (random, unpredictable fluctuations). The goal of a forecasting model is to capture trend and seasonality so that only noise remains in the residuals.

PRACTICE P1 · TIME SERIES FUNDAMENTALS

(a) **Identify the component.** For each observation, classify it as Trend, Seasonality, or Noise.

Observation	Component
Sales of umbrellas spike every monsoon season.	Seasonality
A country's GDP has grown steadily over the past 30 years.	Trend
An unexpected factory fire disrupts production for one week.	Noise
Electricity demand peaks every afternoon and dips every night.	Seasonality
Global average temperature has risen over the past century.	Trend
A sudden viral social media post causes a one-day sales spike.	Noise

Expected answer: Expected Reasoning: Students should distinguish between regular patterns (seasonality), long-term direction (trend), and one-off unpredictable events (noise).

Expected answer: Common Misconception: Students may classify the factory fire as seasonality because it lasted a week. Emphasize that noise is defined by its unpredictability and non-repeating nature, not its duration.

Expected answer: Marking Hint: Award 1 mark per correct classification. Accept "residual" or "random shock" in place of "noise."

(b) True or False. Justify each answer.

Statement	T / F	Justification
Shuffling the rows of a time series dataset does not affect model predictions.	[F]	Shuffling destroys temporal dependencies. Time series models rely on the order of observations; randomizing them removes the sequential structure.
Classical time series models generally assume equally spaced time intervals.	[T]	Most classical time series models require consistent intervals (e.g., every hour, every day). Irregular spacing requires special handling.
Standard K-Fold cross-validation can be used for time series without modification.	[F]	K-Fold randomly assigns data to folds, which can place future data in training and past data in testing, violating temporal ordering and causing data leakage.
If a model's residuals show a repeating pattern, the model has captured all seasonality.	[F]	A repeating pattern in residuals means the model has FAILED to capture the seasonality. If all seasonality were captured, residuals would be pure random noise.

Expected answer: Expected Reasoning: The last item is the most challenging. Students must understand that patterns in residuals indicate unmodeled structure.

Expected answer: Common Misconception: Students may think repeating residuals confirm the model works. The opposite is true — residuals should be structureless noise.

Expected answer: Marking Hint: Award 1 mark for T/F, 1 mark for justification.

(c) Short reasoning. Why is a house price dataset considered cross-sectional but a daily stock price dataset considered a time series? Explain in terms of the I.I.D. assumption.

In a house price dataset, each row is an independent house — the price of a house in Delhi does not depend on the price of a house in Mumbai. The rows satisfy the I.I.D. assumption, so order does not matter. In a stock price dataset, today's price is directly influenced by yesterday's price (temporal dependence). The I.I.D. assumption is violated because consecutive observations are not independent. The temporal ordering carries critical information for prediction.

Expected answer: Expected Reasoning: Students should explicitly mention I.I.D., independence between rows, and temporal dependence.

Expected answer: Common Misconception: Students may focus only on "time" as the distinguishing factor rather than the violation of independence.

Expected answer: Marking Hint: Full marks require mentioning I.I.D., explaining why house data satisfies it, and why stock data violates it.

PART II: MAKING DATA READY FOR FORECASTING

Classical time series models require the data to have stable statistical properties. This section introduces stationarity and the techniques to achieve it.

Stationarity

▷ HOOK

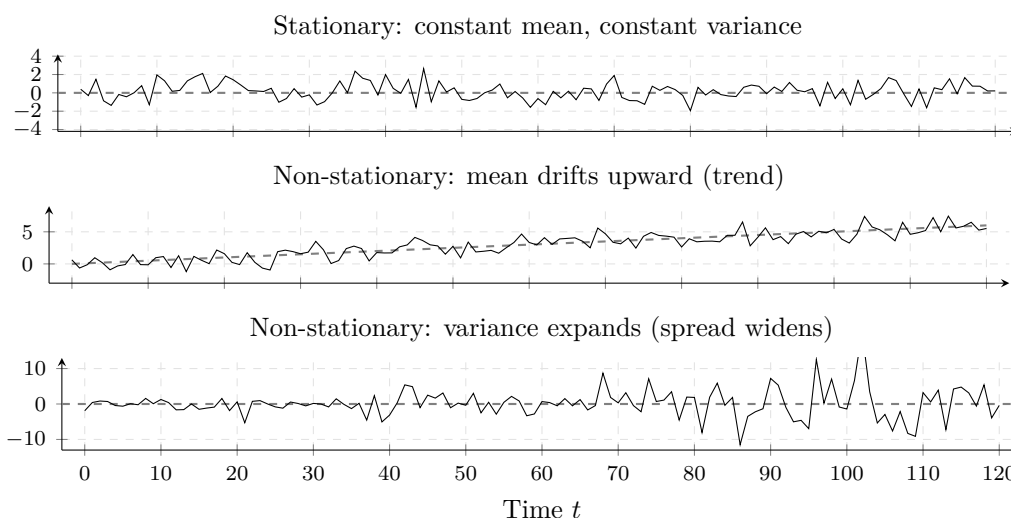
A tech startup's revenue has been growing by exactly 20% every year for the past decade. You want to predict next year's revenue. Can you simply take the arithmetic mean of the last ten years of revenue? The average would be far lower than the current reality because the baseline is constantly moving upward. For our forecasting models to work, the statistical properties of the data must remain stable over time.

Stationary vs. non-stationary series

A **Stationary Time Series** is one whose statistical properties do not change over time. Specifically:

Property	Requirement	In plain language
Mean	Constant	Data fluctuates around a flat, horizontal line
Variance	Constant	The spread of fluctuations stays the same over time
Autocovariance	Constant	The relationship between observations depends on the lag, not on the particular point in time

A **Non-Stationary Time Series** is one where the rules are changing. The mean is **drifting** (trend), or the variance is **expanding** (like a megaphone shape on a graph).



Classroom question: Why do Machine Learning algorithms struggle with non-stationary data?

Expected answer: Machine Learning algorithms are pattern-matching engines. If the rules governing the data are constantly changing (non-stationary), the algorithm cannot reliably learn from the past to predict the future. Patterns learned from historical data may not apply to future data because the underlying distribution has shifted. It is like trying to learn the rules of chess while someone keeps changing how the pieces move.

★ KEY INSIGHT

Machine Learning algorithms are pattern-matching engines. If the rules governing the data are constantly changing (non-stationary), the algorithm cannot reliably learn from the past to predict the future. Stationarity ensures that patterns learned from historical data remain valid for future predictions.

◆ TAKEAWAY

Classical time series models require stationary data — data with constant mean, constant variance, and constant autocovariance. Most real-world data is non-stationary, so it must be transformed before modeling.

Lags: the features of time series

▷ HOOK

In standard regression, we predict house price using features like area, bedrooms, and location. But in time series, we have only one column: the past values of our target variable. If you want to predict today's maximum temperature and are allowed only one piece of historical data, what would you ask for? Most likely, yesterday's temperature. This past observation — called a lag — becomes our feature.

Defining lags

Notation	Name	Meaning
y_t	Current observation	Today's value
y_{t-1}	Lag 1	Yesterday's value
y_{t-2}	Lag 2	Day before yesterday's value
y_{t-7}	Lag 7	Same day last week (weekly seasonality)
y_{t-k}	Lag k	Value k time steps ago

By defining lags, we solve a major engineering problem: we represent the past mathematically as distinct variables. This allows us to feed historical information into an equation just like we feed features into a regression model.

◆ TAKEAWAY

Lags are the features of time series. They convert the temporal structure into a format that regression-style models can use. Lag 1 is the most recent past observation; Lag k is the observation k steps ago.

Making a series stationary: Differencing

▷ HOOK

We have a contradiction: our models require stationary data, but real-world data typically has strong trends and seasonality. How do we bridge this gap? Instead of predicting the absolute value, we predict the change from one time step to the next.

Lag differencing

If the data has a trend (constantly moving), we transform it by subtracting the previous observation from the current one. This technique is called **Lag Differencing**.

$$\Delta Y_t = Y_t - Y_{t-1}$$

Symbol	Meaning
ΔY_t	The new, differenced value at time t
y_t	The original, raw observation at time t
y_{t-1}	The original observation at the previous time step

Numerical example

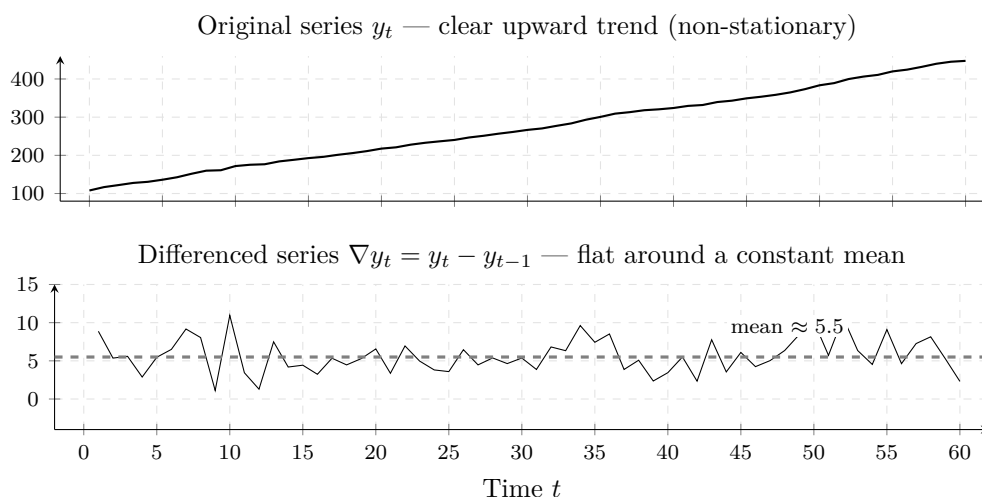
Time (t)	Raw Stock Price y_t	Differenced Series ΔY_t
Day 1	100	— (no prior day)
Day 2	105	105 – 100 = 5
Day 3	110	110 – 105 = 5
Day 4	115	115 – 110 = 5
Day 5	122	122 – 115 = 7

The original data (100, 105, 110, 115, 122) has a clear **upward** trend. The differenced data fluctuates around a roughly constant mean, making it **stationary**.

Expected answer: Expected Reasoning: Students should perform the subtraction and observe that the differenced series (5, 5, 5, 7) has a roughly constant mean (5.5) with no trend.

Expected answer: Common Misconception: Students may think differencing removes seasonality. First-order differencing primarily removes trend. Seasonal differencing (subtracting the value from the same season last year) is needed for seasonality.

Expected answer: Marking Hint: Award marks for correct arithmetic and correct identification of the stationarity transformation.



★ KEY INSIGHT

Differencing removes the trend by converting absolute values into period-to-period changes. The resulting series fluctuates around a constant mean, satisfying the stationarity requirement.

◆ TAKEAWAY

Lag differencing ($\Delta Y_t = Y_t - Y_{t-1}$) transforms non-stationary data into stationary data by removing the trend. Instead of modeling the raw values, we model the changes between consecutive time steps.

PRACTICE P2 · STATIONARITY, DIFFERENCING, AND LAGS

(a) **Diagnose stationarity.** For each description, state whether the series is stationary or non-stationary and identify the violated property.

Description	Stationary?	Which property is violated?
Data fluctuates randomly around a constant value of 50 with similar spread throughout.	Stationary	None — all properties satisfied
Sales show a steady upward climb over five years.	Non-stationary	Constant mean (upward trend)
Stock returns oscillate around zero, but the swings become wider each year.	Non-stationary	Constant variance (expanding spread)
Hourly temperature readings show no trend and similar variation all year.	Stationary	None — all properties satisfied

Expected answer: Expected Reasoning: Students should check each of the three stationarity conditions and identify which specific one is violated.

Expected answer: Common Misconception: Students may say the temperature data is non-stationary because temperature changes throughout the day. Clarify that if the daily pattern repeats consistently with similar mean and variance, the statistical properties are stable.

Expected answer: Marking Hint: Award marks for correct stationarity classification AND correct identification of the violated property.

(b) Apply differencing. Complete the table.

Month	Monthly Sales y_t	Differenced $\Delta Y_t = Y_t - Y_{t-1}$
Jan	200	—
Feb	220	20
Mar	250	30
Apr	245	−5
May	270	25
Jun	300	30

(i) Does the original series appear stationary? **No.** Why?

The values increase from 200 to 300 over six months, showing an upward trend. The mean is not constant.

(ii) Does the differenced series appear more stationary? **Yes.** Why?

The differenced values (20, 30, −5, 25, 30) fluctuate around a roughly constant mean (20) without a clear trend. The stationarity of the mean has been improved.

Expected answer: Expected Reasoning: Students should compute each difference correctly and observe the mean behavior in both series.

Expected answer: Common Misconception: Students may claim the differenced series is perfectly stationary. It appears more stationary, but a formal statistical test (like ADF) would be needed to confirm.

Expected answer: Marking Hint: Award marks for correct arithmetic (5 values) and correct qualitative reasoning.

(c) Identify the lag. Express each statement using lag notation.

Statement	Lag notation
"The temperature two days ago"	y_{t-2}
"Last week's sales (weekly data)"	y_{t-1} or y_{t-7} if daily data
"The observation three time steps ago"	y_{t-3}

Expected answer: Expected Reasoning: Direct application of lag notation.

Expected answer: Common Misconception: For "last week's sales," students may write y_{t-7} assuming daily data. If the data is weekly, last week is simply y_{t-1} . Accept either with justification.

Expected answer: Marking Hint: Award marks for correct notation. Accept y_{t-7} for the weekly question if the student specifies daily frequency.

(d) Assertion–Reason.

	Statement	True / False
Assertion	Differencing is used to make a non-stationary series stationary.	[T]
Reason	Differencing removes the trend by converting absolute values into period-to-period changes.	[T]

The Reason is a correct explanation of the Assertion: [T]

Expected answer: Expected Reasoning: Both are true and the Reason directly explains the mechanism behind the Assertion.

Expected answer: Common Misconception: Students may think differencing removes seasonality as well. First-order differencing removes trend; seasonal differencing is a separate step.

Expected answer: Marking Hint: Award 1 mark each for Assertion, Reason, and relationship.

PART III: BUILDING FORECASTING MODELS

With stationary data and lag features defined, we can now build models that predict the future using the past.

Auto Regression (AR) models

▷ HOOK

Recall Multiple Linear Regression: we predicted a target Y using feature columns X_1, X_2, X_3 . But in time series, we do not have separate feature columns like "area" or "bedrooms." We have only one column — the past values of our target variable. What if we treat our lags exactly like the features in a regression model? We would be running a regression of a variable against historical versions of itself.

The AR model equation

The **Auto Regression (AR)** model predicts the current value as a weighted sum of its own past values. The prefix "Auto" means **self**. We are running a regression of a variable against **historical versions of itself**.

The Auto Regression equation of order p , denoted $AR(p)$:

$$y_t = c + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \cdots + \phi_p \cdot y_{t-p} + \varepsilon_t$$

Symbol	Meaning
y_t	The current value we are trying to predict today
c	A constant intercept (the baseline mean of the series)
y_{t-1}	Lag 1 — yesterday's value, acting as Feature 1
ϕ_1	The coefficient (weight) assigned to Lag 1
p	The order of the model — how many lags we look back
ε_t	White noise — the unpredictable error term for today

Example: An $AR(2)$ model for daily temperature prediction would be:

$$y_t = c + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \varepsilon_t$$

This uses **two** lags (yesterday and the day before yesterday) as features.

★ KEY INSIGHT

The AR model operates on a principle of **momentum**: if the data had high values yesterday and the day before, the positive weights (ϕ) carry that momentum forward to predict a high value for today. It transforms the problem of predicting over time into a standard weighted sum — like Linear Regression where the features are the variable's own past values.

The problem of choosing p

Classroom question: What happens if we set $p = 365$, forcing the model to learn 365 weights for predicting tomorrow's temperature?

Expected answer: The model will become overly complex, slow, and will likely overfit by learning useless noise from a random day 8 months ago. This is the same overfitting problem from Lecture 9. Too many parameters relative to the data means the model memorizes training noise rather than learning genuine patterns.

Setting p too **large** leads to overfitting. Setting p too **small** may miss important patterns. We need a mathematical tool to determine which past lags carry statistically significant signals.

◆ TAKEAWAY

The $AR(p)$ model predicts the current value as a linear combination of p past values plus an error term. It converts time series forecasting into a regression problem where lags serve as features. The critical question is choosing the right number of lags p .

Autocorrelation (ACF) and Partial Autocorrelation (PACF)

▷ HOOK

We need to select p — the number of lags for our AR model — but we cannot just guess. We need to objectively measure the strength of the relationship between observations separated by different time gaps. You already know Pearson correlation, which measures the relationship between two different variables. What if we apply the same idea, but between a variable and a lagged version of itself?

Autocorrelation Function (ACF)

Autocorrelation measures the linear relationship between a variable and a **lagged version** of itself.

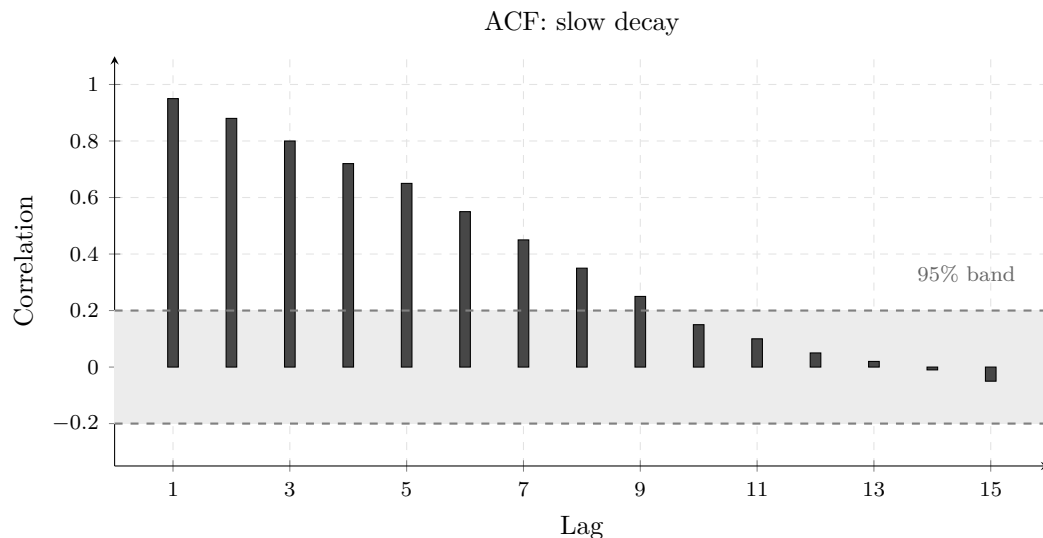
The ACF at lag k calculates the correlation between the original series y_t and the shifted series y_{t-k} .

The formula for ACF at lag k :

$$\rho_k = \frac{\sum_{t=k+1}^N (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^N (y_t - \bar{y})^2}$$

where $\bar{y}$ is the mean of the series and N is the total number of observations.

ACF value	Interpretation
Near +1	Strong positive correlation: when the past went up, today tends to go up (trend momentum)
Near -1	Strong negative correlation: when the past went up, today tends to go down (mean-reverting behavior)
Near 0	No linear relationship: that specific past lag offers no useful information for forecasting today



The problem with ACF: indirect correlations

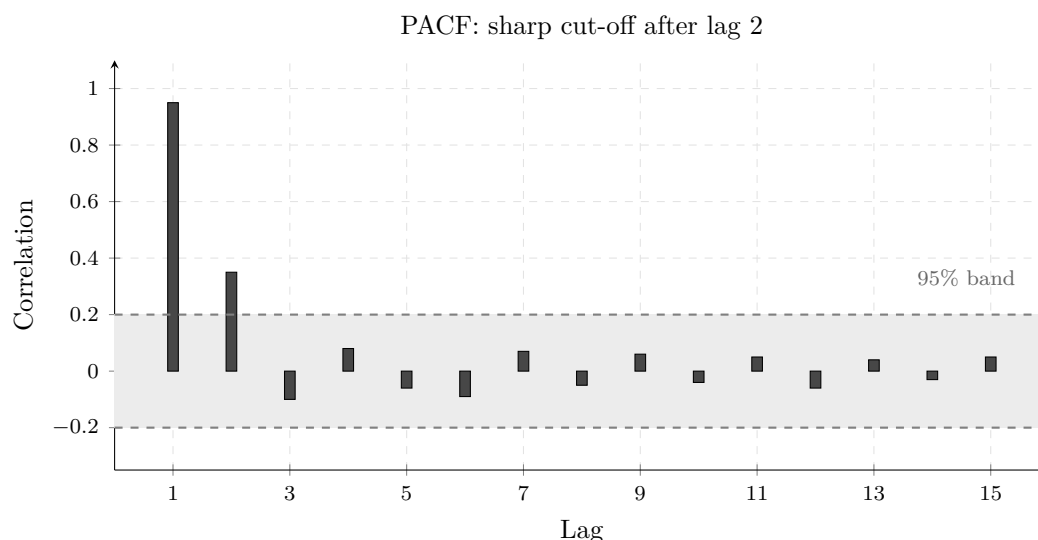
Classroom question: Today's temperature is highly correlated with yesterday's. Yesterday's was highly correlated with the day before. Standard ACF will therefore show that today is correlated with the day before yesterday. Is that a direct relationship, or a ripple effect passing through yesterday?

Expected answer: It is mostly a ripple effect. The day before yesterday influences today primarily by influencing yesterday first. The correlation between today and two days ago is largely indirect — it passes through yesterday as an intermediary.

Standard ACF captures the **total** effect (both direct and indirect). It might tell us we need 10 lags when in reality we only need 1 lag carrying a chain reaction forward.

Partial Autocorrelation Function (PACF)

PACF measures the pure, **direct** correlation between y_t and y_{t-k} after completely removing the influence of all **intermediate** lags in between. It isolates the unique new information that lag k brings.



ACF decays slowly — lag 1 correlation ripples through to lag 2, lag 3 and beyond. PACF strips those ripples out and cuts off sharply after lag 2, revealing only the direct dependencies.

How to read a PACF plot to choose p

1. Look at the PACF plot.
2. Identify the shaded confidence band (the significance threshold).
3. Count how many lag bars extend significantly **beyond** the band before they drop into the noise.
4. If only Lag 1 and Lag 2 are significant, set $p = 2$. Build an AR(2) model.

★ KEY INSIGHT

ACF shows the total correlation (direct + indirect) between a series and its lags. PACF removes indirect effects and shows only the direct correlation. To choose p for an AR model, count the number of significant bars in the PACF plot.

Tool	What it measures	Use for
ACF	Total correlation (direct + indirect)	Determining the order q of an MA model (covered next)
PACF	Direct correlation only	Determining the order p of an AR model

◆ TAKEAWAY

ACF measures the total linear relationship between a series and its lags, but includes indirect ripple effects. PACF isolates the direct effect of each lag by removing intermediate influences. Use the PACF plot to determine the order p of an AR model.

PRACTICE P3 · AUTO REGRESSION, ACF, AND PACF

(a) **Write the AR equation.** Write out the full AR(3) equation for predicting daily sales, labeling each term.

$$y_t = c + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \varepsilon_t$$

Expected answer: Expected Reasoning: Students should substitute $p = 3$ and write three

lag terms with distinct coefficients.

Expected answer: Common Misconception: Students may forget the constant c or the error term ε_t , or use the same coefficient for all lags.

Expected answer: Marking Hint: Award marks for correct structure: constant + 3 weighted lag terms + error. Deduct for missing c or ε_t .

(b) Interpret a PACF plot. A PACF plot for a temperature dataset shows the following:

Lag	1	2	3	4	5	6	7
PACF value	0.85	0.42	0.05	−0.02	0.03	−0.01	0.38
Significant?	Yes	Yes	No	No	No	No	Yes

(i) Based on lags 1–6 only, what order AR model would you initially choose? $p = 2$ (only lags 1 and 2 are significant among the first 6).

(ii) Lag 7 is also significant. What real-world pattern might this represent?

A weekly seasonality pattern. If the data is daily, lag 7 corresponds to the same day of the previous week. Temperature or activity patterns often repeat on a weekly cycle (e.g., weekday vs. weekend).

(iii) Would you include lag 7 in your model? Why or why not?

Yes, it should likely be included because it captures a meaningful weekly seasonal pattern. The model could be built with lags 1, 2, and 7 as features (though this is no longer a simple AR(2) — it would require either an AR(7) or a model with specific lag selection).

Expected answer: Expected Reasoning: Students should recognize that lag 7 in daily data suggests weekly seasonality, which is a real pattern rather than noise.

Expected answer: Common Misconception: Students may think we must include all lags up to the last significant one (i.e., AR(7) with lags 3–6 included). In practice, specific lag selection can be used.

Expected answer: Marking Hint: Award marks for correctly identifying $p = 2$ from the first 6 lags, recognizing the weekly pattern, and reasoning about inclusion.

(c) Compare ACF and PACF. Fill in the blanks.

Property	ACF	PACF
Measures _____ correlation	Total (direct + indirect)	Direct only
Behavior for an AR process	Decays slowly (tails off)	Cuts off sharply after p lags
Used to determine	Order q of an MA model	Order p of the AR model

Expected answer: Expected Reasoning: Students should recall that ACF includes ripple effects (decays slowly) while PACF isolates direct effects (cuts off sharply).

Expected answer: Common Misconception: Students may reverse the roles — thinking ACF determines p and PACF determines q . Emphasize: PACF $\rightarrow$ AR order; ACF $\rightarrow$ MA order.

Expected answer: Marking Hint: Award marks for all blanks correct. The reversal of ACF/PACF roles is the most critical error to check.

(d) MCQ. An AR(1) model with $\phi_1 = 0.9$ and $c = 2$ predicts y_t . If $y_{t-1} = 50$, what is the predicted value (ignoring the error term)?

(A) 47 (B) 45 (C) 50 (D) 52

Answer: **(A) 47** — substituting into the AR(1) equation, $y_t = c + \phi_1 \cdot y_{t-1} = 2 + 0.9 \times 50 = 2 + 45 = 47$.

Expected answer: Expected Reasoning: Direct substitution into the AR(1) equation.

Expected answer: Common Misconception: Students may forget to add the constant c , getting 45 instead of 47.

Expected answer: Marking Hint: Award full marks for (A). If a student selects (B) = 45, they forgot the constant.

(e) Error identification. A student writes: "To choose p for an AR model, I look at the ACF plot and count the significant bars." Identify and correct the error.

The error is using the ACF plot instead of the PACF plot. ACF includes both direct and indirect correlations (ripple effects), so it will overestimate the number of necessary lags. The correct approach is to use the PACF plot, which isolates the direct correlation of each lag after removing intermediate effects. PACF bars that are significant indicate the true, direct dependencies needed for the AR model.

Expected answer: Expected Reasoning: Students should identify the specific tool confusion and explain why PACF is preferred.

Expected answer: Common Misconception: Students may think ACF and PACF are interchangeable. They are not — each serves a different purpose.

Expected answer: Marking Hint: Full marks require identifying ACF as the wrong tool, naming PACF as correct, and explaining the difference (direct vs. total correlation).

PART IV: MODELING EXTERNAL SHOCKS

The AR model assumes the future is driven entirely by the momentum of past values. But real-world systems are constantly hit by unpredictable external events.

Random shocks and Moving Average (MA) models

▷ HOOK

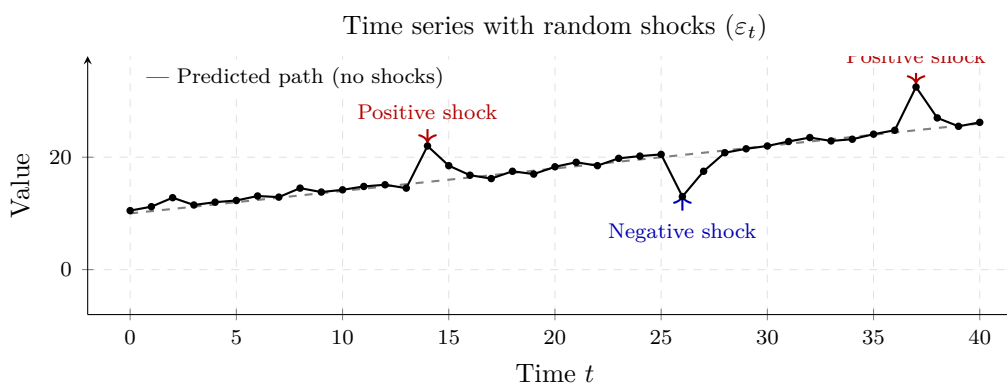
Your AR model is predicting daily stock prices based on yesterday's and the day before's values. At 2:00 PM today, a sudden government policy announcement causes a massive market crash. Did yesterday's stock price predict that policy announcement? No — it was completely unpredictable from the historical data. These external, unpredictable events constantly push time series off their natural paths.

Random shocks (innovations)

In Time Series Analysis, unpredictable external events are called **Random Shocks** (or **innovations**). Mathematically, random shocks are the **forecast errors** at each time step:

$$\text{Random Shock} = \text{Actual value} - \text{Predicted value} = \varepsilon_t$$

This is the ε_t term that appears in the AR equation.



The dashed gray line shows the predicted path. Random shocks (ε_t) push the actual values above or below the prediction at unpredictable times.

The Moving Average (MA) model

What if we build a model that learns from the shocks themselves, rather than just the past values?

Important: Do NOT confuse this with a "rolling mean" (like a 50-day moving average on a stock chart). The MA model in forecasting uses **past forecast errors (random shocks)** as its predictors.

The Moving Average equation of order q , denoted $MA(q)$:

$$y_t = c + \varepsilon_t + \theta_1 \cdot \varepsilon_{t-1} + \theta_2 \cdot \varepsilon_{t-2} + \cdots + \theta_q \cdot \varepsilon_{t-q}$$

Symbol	Meaning
y_t	The value we are trying to predict today
c	The constant mean of the series
ε_t	The current random shock happening right now (unpredictable)
ε_{t-1}	The random shock (forecast error) that happened yesterday
θ_1	The coefficient/weight assigned to yesterday's shock
q	The order of the MA model — how many past shocks we look at

AR vs. MA: two fundamentally different assumptions

Aspect	AR (Auto Regression)	MA (Moving Average)
Predicts using	Past values $y_{t-1}, y_{t-2}, \dots$	Past errors $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$
Assumes current values are driven by	SIGNAL of the past	NOISE of the past
Intuition	Momentum: if sales were high yesterday, they will be high today	Shock absorption: a temporary bump slowly fades over q days
Order parameter	p (number of lags)	q (number of past shocks)

Expected answer: Expected Reasoning: Students should understand the fundamental philosophical difference: AR uses past values (momentum/signal), MA uses past errors (shock absorption/noise).

Expected answer: Common Misconception: Students frequently confuse the MA model with a rolling/moving average calculation. The MA model in forecasting is fundamentally different — it uses past forecast errors, not past values.

Expected answer: Marking Hint: Check that students distinguish between past values and past errors. This is the most critical conceptual distinction in this section.

★ KEY INSIGHT

AR and MA models capture two fundamentally different dynamics. AR models assume **momentum** — if sales were high yesterday, they will naturally be high today. MA models assume the baseline is mostly flat, but temporary **shocks** (like a good marketing campaign) bump the series off its baseline, and that bump slowly fades over q time steps.

◆ TAKEAWAY

Random shocks are the unpredictable, external events that push a time series off its natural path. The $MA(q)$ model learns from these past forecast errors. AR models use past values (signal); MA models use past errors (noise). In the next lecture, combining both leads to the powerful ARMA and ARIMA models.

PRACTICE P4 · COMPREHENSIVE CHALLENGE (Homework)

(a) Fill in the Key Terms table.

Term	Definition
Time Series	A sequence of data points recorded at successive, equally spaced intervals in time.
Stationarity	A property where the statistical properties (mean, variance, autocovariance) of a series do not change over time.
Differencing	A transformation that subtracts the previous observation from the current one ($\Delta Y_t = Y_t - Y_{t-1}$) to remove trend and achieve stationarity.
Lag	A past observation at a specific time delay; Lag k refers to y_{t-k} , the value k time steps ago.
Autocorrelation (ACF)	The correlation between a time series and a lagged version of itself, capturing both direct and indirect relationships.
Partial Autocorrelation (PACF)	The direct correlation between y_t and y_{t-k} after removing the influence of all intermediate lags.
AR(p)	Auto Regression model of order p — predicts the current value as a weighted sum of p past values plus an error term.
MA(q)	Moving Average model of order q — predicts the current value using the mean plus a weighted sum of q past forecast errors.
Random Shock	An unpredictable external event that causes the actual value to differ from the model's prediction; mathematically, the forecast error ε_t .

Expected answer: Expected Reasoning: Definitions should be precise and include the mathematical form where applicable.

Expected answer: Common Misconception: Students may define MA as "average of past values." It uses past forecast errors, not past values.

Expected answer: Marking Hint: Award marks for accurate, concise definitions. Deduct for confusing MA with rolling average.

(b) Match the columns.

Concept	Description
(1) Trend	(A) The correlation of a series with a lagged version of itself
(2) Seasonality	(B) An unpredictable external event that causes forecast error
(3) Noise	(C) Short-term repeating cycles at fixed frequencies
(4) Stationarity	(D) A series whose statistical properties do not change over time
(5) ACF	(E) Random, unexplainable variation after trend and seasonality are removed
(6) Random Shock	(F) The long-term direction of the data

Matches: (1)–F, (2)–C, (3)–E, (4)–D, (5)–A, (6)–B

Expected answer: Expected Reasoning: Direct matching of definitions to concepts.

Expected answer: Common Misconception: Students may confuse Noise (E) with Random Shock (B). Noise is the general residual component; a random shock is a specific unpredictable event.

Expected answer: Marking Hint: Award 1 mark per correct match.

(c) Case study. A retail store records weekly sales data for 3 years. The data shows a steady upward trend and a spike every December (holiday season).

(i) Is this data stationary? **No.** Identify the non-stationary components.

The data has an upward trend (changing mean) and seasonal spikes every December (repeating pattern). Both violate the constant mean requirement.

(ii) What transformation would you apply first to address the trend?

First-order differencing: $\Delta Y_t = Y_t - Y_{t-1}$. **This removes the upward trend by converting absolute sales into week-to-week changes.**

(iii) After making the data stationary, the PACF plot shows significant bars at lags 1 and 2 only. What model would you build?

An AR(2) model: $y_t = c + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \varepsilon_t$. **The PACF indicates that only the two most recent weeks have a direct influence on the current week.**

(iv) After fitting the model, the residuals show a repeating weekly pattern. What does this indicate?

The model has failed to capture all the structure in the data. A repeating pattern in residuals means there is unexplained seasonality. The model should be improved — perhaps by including additional seasonal lags or using a different model that accounts for seasonality.

Expected answer: Expected Reasoning: This ties together multiple concepts: stationarity diagnosis, differencing, PACF interpretation, AR model selection, and residual analysis.

Expected answer: Common Misconception: Students may think repeating residuals are acceptable. Ideal residuals should be pure noise with no discernible pattern.

Expected answer: Marking Hint: Award marks for each part. Part (iv) is the most challenging — full marks require recognizing that patterned residuals indicate model failure.

(d) Compare and contrast. Complete the table.

Aspect	AR model	MA model
Inputs to the model	Past values of the series $y_{t-1}, y_{t-2}, \dots$	Past forecast errors $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$
What it assumes drives current values	The signal/momentum of past values	The noise/shocks from the past
Order parameter name	p — number of past values (lags)	q — number of past shocks (errors)
Determined using	PACF plot	ACF plot

Expected answer: Expected Reasoning: Each row tests a distinct conceptual difference. The last row is especially important.

Expected answer: Common Misconception: Students may reverse which plot determines which model's order. PACF $\rightarrow$ AR; ACF $\rightarrow$ MA.

Expected answer: Marking Hint: Award 1 mark per correct cell. The last row is worth extra emphasis.

(e) **Synthesis questions.** Answer any three.

(i) Why does shuffling the rows of a time series dataset destroy useful information, but shuffling a cross-sectional dataset does not?

In cross-sectional data, each row is an independent observation — its meaning does not depend on neighboring rows. In time series, the temporal ordering encodes critical dependency information: today's value is influenced by yesterday's value, and yesterday's by the day before. Shuffling breaks these sequential dependencies, converting a meaningful signal into random noise. The I.I.D. assumption holds for cross-sectional data but not for time series.

(ii) Explain the relationship between stationarity and the reliability of forecasting models.

Forecasting models learn patterns from historical data and apply them to predict the future. If the data is stationary (constant mean, variance, and autocovariance), patterns learned from the past remain valid for the future. If the data is non-stationary, the statistical properties shift over time, meaning rules learned from older data may not apply to newer data. Stationarity ensures that the model's learned rules remain stable and transferable to future predictions.

(iii) Why is PACF more useful than ACF for selecting the order of an AR model?

ACF captures the total correlation between a series and its lags, including indirect effects that pass through intermediate time steps (chain reactions). This causes ACF to show many significant lags even when only one or two direct dependencies exist. PACF mathematically removes these indirect effects and shows only the direct correlation of each lag. This allows us to accurately count how many lags truly have a direct influence, giving us the correct order p for the AR model.

(iv) A friend says: "The MA model uses a rolling average of past data to predict the future." Is this correct? Explain.

This is incorrect. The MA model in forecasting does NOT use a rolling average of past values. Instead, it uses past forecast errors (random shocks) as predictors. The "Moving Average" name is misleading — it refers to a weighted sum of past error terms $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$, not a rolling mean of $y_{t-1}, y_{t-2}, \dots$. The rolling average (like a 50-day moving average on a stock chart) is a completely different concept.

(v) How does the concept of lags connect time series forecasting to Linear Regression?

Lags transform the time series problem into a standard regression problem. By defining past values $y_{t-1}, y_{t-2}, \dots, y_{t-p}$ as "features" (like area, bedrooms in house price prediction), we can write the AR model as a Linear Regression equation:

$y = c + \phi_1 X_1 + \phi_2 X_2 + \dots + \phi_p X_p + \varepsilon$, where each X_j is a lag of the target variable. This is why AR models are called "regression of a variable against itself."

(vi) Why is differencing necessary before applying an AR model to data with an upward trend?

An AR model assumes stationarity — that the statistical properties of the data remain constant over time. An upward trend violates the constant mean requirement because the average value keeps increasing. If we fit an AR model to non-stationary data, the model will produce unreliable predictions because the patterns it learned from past data (centered around a lower mean) will not apply to future data (centered around a higher mean). Differencing removes the trend, creating a stationary series that the AR model can reliably learn from.

Expected answer: Expected Reasoning: These require synthesis of multiple concepts. Full-credit answers should demonstrate understanding of the mechanism, not just the outcome.

Expected answer: Common Misconception: For (iv), the rolling average confusion is extremely common and must be addressed directly.

Expected answer: Marking Hint: Award 3 marks per question (any three). Require both the what and the why for full credit.

BY THE END OF THIS SHEET, YOU CAN

- ☒ Explain why time series data violates the I.I.D. assumption and why row order matters.
- ☒ Decompose an observed time series into Trend, Seasonality, and Noise components.
- ☒ Define stationarity and identify whether a given series is stationary or non-stationary.
- ☒ Apply lag differencing to transform a non-stationary series into a stationary one.
- ☒ Define lags and explain how they serve as features for time series models.
- ☒ Write the AR(p) equation and interpret each term.
- ☒ Distinguish between ACF and PACF and use the PACF plot to select the order p of an AR model.
- ☒ Define random shocks and write the MA(q) equation.
- ☒ Compare AR and MA models in terms of inputs, assumptions, and use cases.

Worksheet 13 · AML · Lecture 13: Time Series Analysis.